



Chapter 2

Network Models

2-1 LAYERED TASKS

*We use the concept of **layers** in our daily life. As an example, let us consider two friends who communicate through postal mail. The process of sending a letter to a friend would be complex if there were no services available from the post office.*

Example:

Alice wants to send a letter to Bob who lives in another city. Without the postal service, Alice would have to:

- Travel to Bob's city herself*
- Find his exact location*
- Deliver the letter personally*

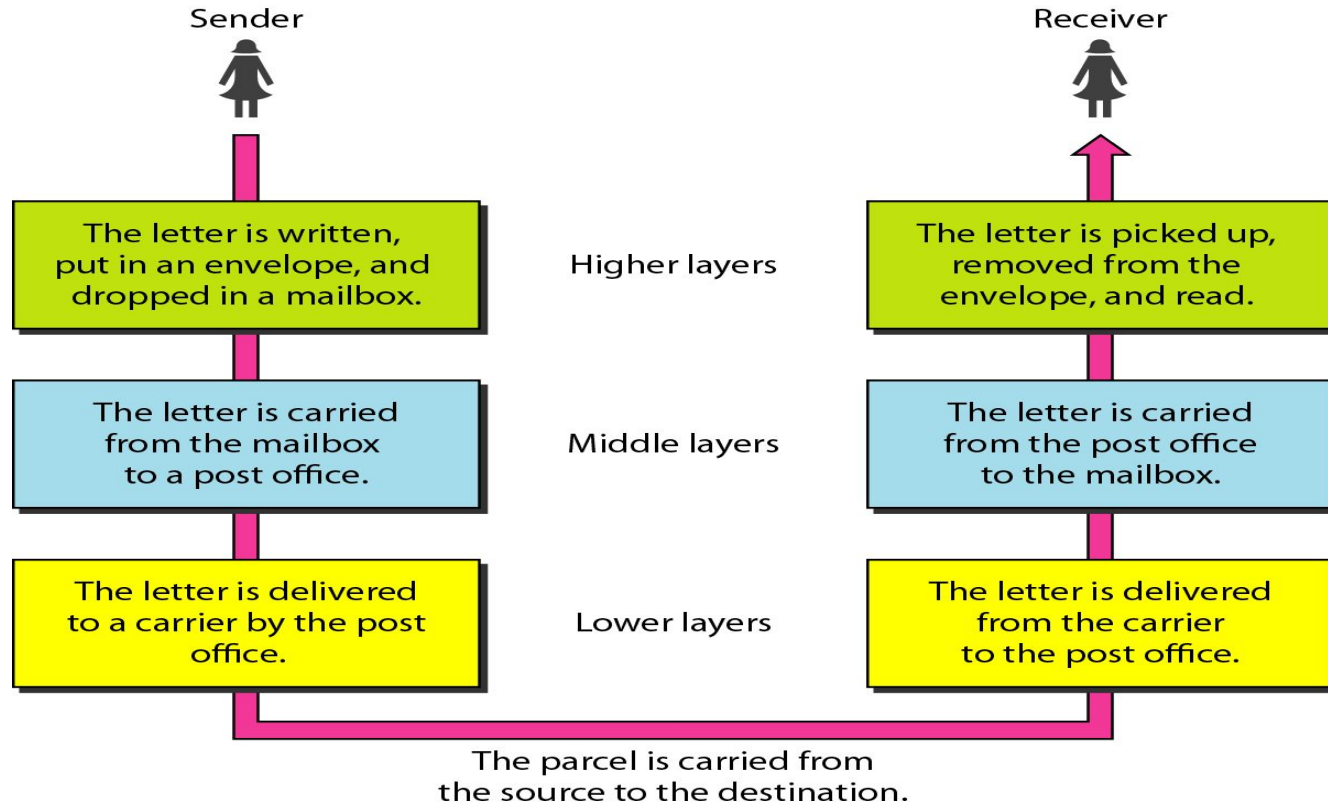
That's a lot of work just like two computers communicating directly without any system or protocol.

2-1 How Layers Help (Postal System Analogy)

To simplify communication, different layers (services) handle different parts of the job:

Layer	Role (Postal Analogy)	Explanation
Application Layer	Writing the letter	Alice writes what she wants to say (message creation).
Presentation Layer	Formatting the letter	She writes in a language both understand and formats it clearly.
Session Layer	Starting and ending communication	She decides when to start and stop sending letters.
Transport Layer	Ensuring delivery	She puts the letter in an envelope with sender/receiver info.
Network Layer	Choosing the route	The postal system decides how to route the letter between cities.
Data Link Layer	Local delivery	The post office in each city ensures it reaches the right area.
Physical Layer	Physical movement	Trucks, airplanes, and mail carriers physically move the letter.

Figure 2.1 Tasks involved in sending a letter



Each layer handles a specific task, and together they make communication smooth and organized — the same concept used in network models like **OSI** or **TCP/IP**.

2-2 THE OSI MODEL

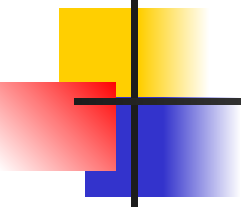
- *Established in 1947, the International Standards Organization (**ISO**) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (**OSI**) model. It was first introduced in the late 1970s.*
- ***OSI** (Open Systems Interconnection) Model is a conceptual framework that describes how data moves from one computer to another over a network.*

Topics discussed in this section:

Layered Architecture

Peer-to-Peer Processes

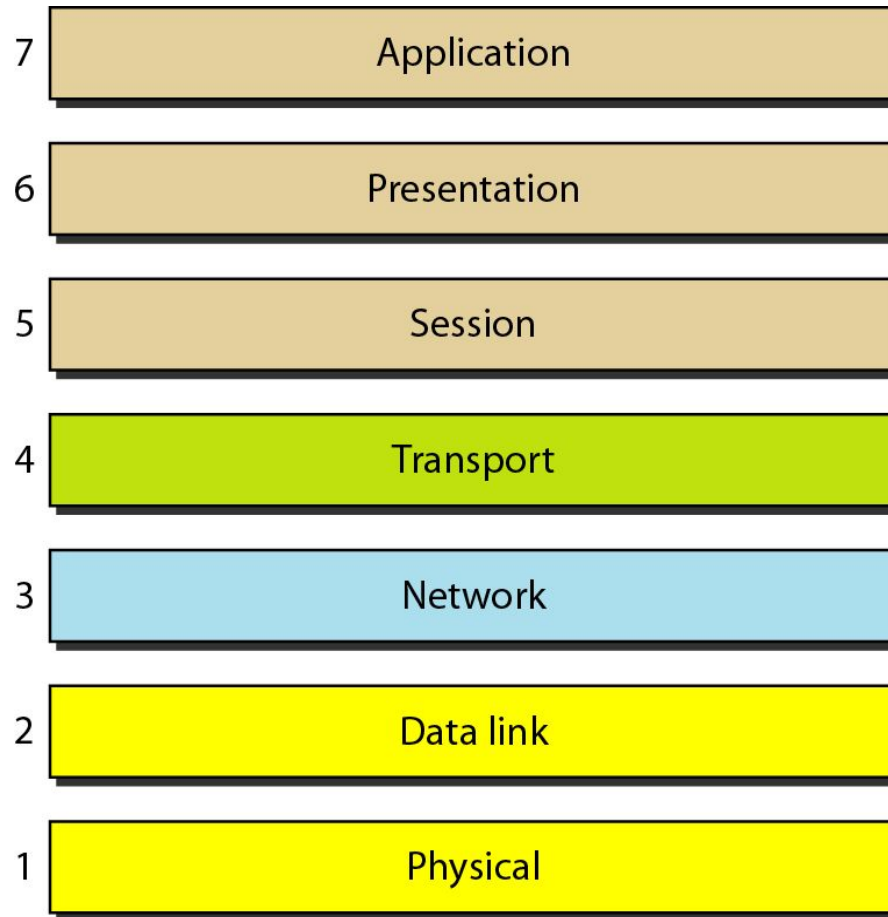
Encapsulation



Note

**ISO is the organization.
OSI is the model.**

Figure 2.2 *Seven layers of the OSI model*



2-3 LAYERS IN THE OSI MODEL

In this section we briefly describe the functions of each layer in the OSI model.

Topics discussed in this section:

- 1. Physical Layer**
- 2. Data Link Layer**
- 3. Network Layer**
- 4. Transport Layer**
- 5. Session Layer**
- 6. Presentation Layer**
- 7. Application Layer**

Figure 2.5 *Physical layer*

From Data Link Layer (Sender Side)

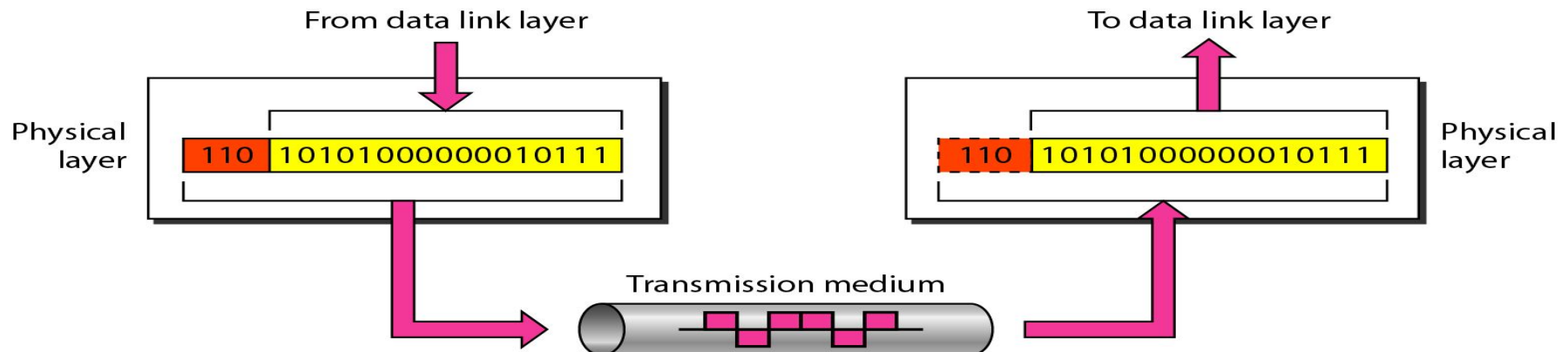
- The data link layer passes a frame (a **sequence of bits**) down to the physical layer.
- This frame (shown in yellow and orange) consists of:
 - **Header bits (110)** — may represent synchronization or control information.
 - **Data bits (10101000000010111)** — the actual payload.

Physical Layer (Sender Side)

- The physical layer converts this digital data into **electrical, optical, or radio signals** suitable for the transmission medium. These signals are then sent through the transmission medium.

Physical Layer (at Receiver Side)

- The receiving physical layer receives the signals and converts them back into bits (0s and 1s). It then passes this bit stream up to the data link layer for further processing.



Note

The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Figure 2.6 *Data link layer*

Sending Process (Encapsulation at the Data Link Layer)

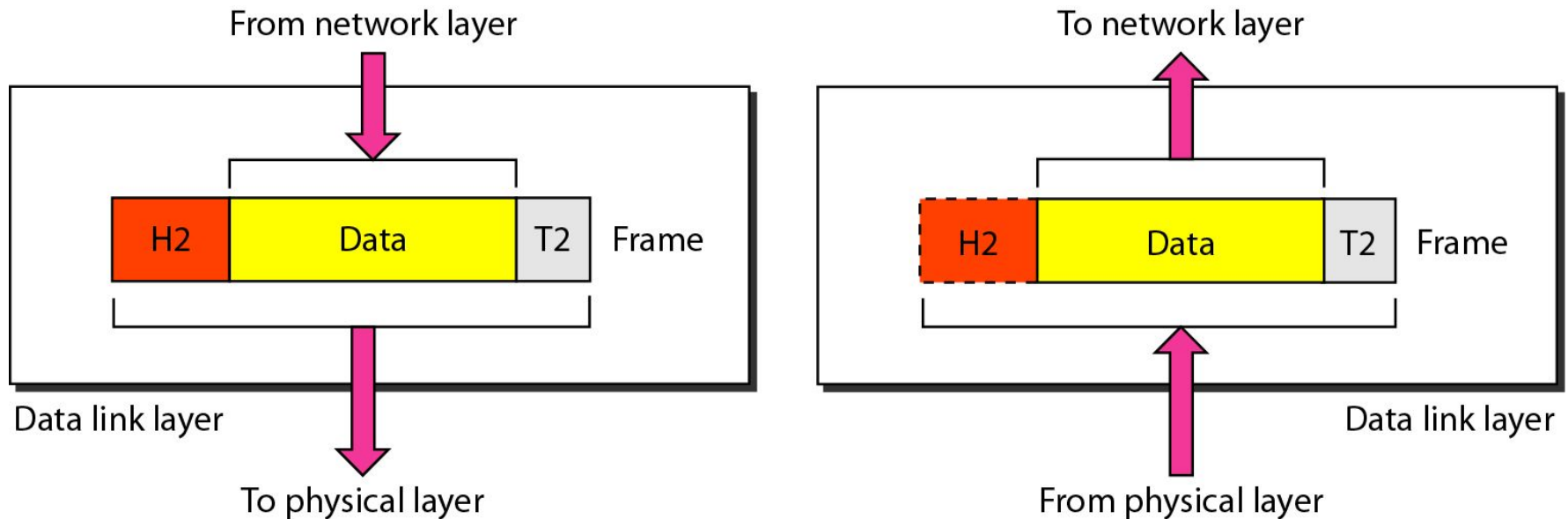
From Network Layer:

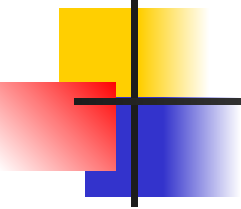
The data link layer receives a packet (called “Data” in yellow) from the network layer above it.

Encapsulation (Framing): The data link layer adds: Header (H2) → contains addressing information like source and destination MAC addresses.

Trailer (T2) → usually contains error detection information (e.g., CRC – Cyclic Redundancy Check). Together, they form a Frame (**H2 + Data + T2**).

At Physical Layer: This frame is then passed down to the physical layer, which converts it into bits (0s and 1s) and sends it across the transmission medium.





Receiving Process (De-encapsulation at the Data Link Layer)

1. From Physical Layer:

- The physical layer at the receiving end delivers the raw bits to the data link layer, which reconstructs the frame.

2. De-encapsulation:

1. The data link layer checks the trailer (T2) for errors.
2. It removes the header (H2) and trailer (T2) after verifying correctness.
3. Only the original data is passed up to the network layer.

Note

The data link layer is responsible for moving frames from one hop (node) to the next.

Figure 2.7 Hop-to-hop delivery

Hop-to-hop delivery means data travels from one node (device) to the next directly connected node not all the way to the destination yet. The Data Link Layer and Physical Layer work together to make this possible.

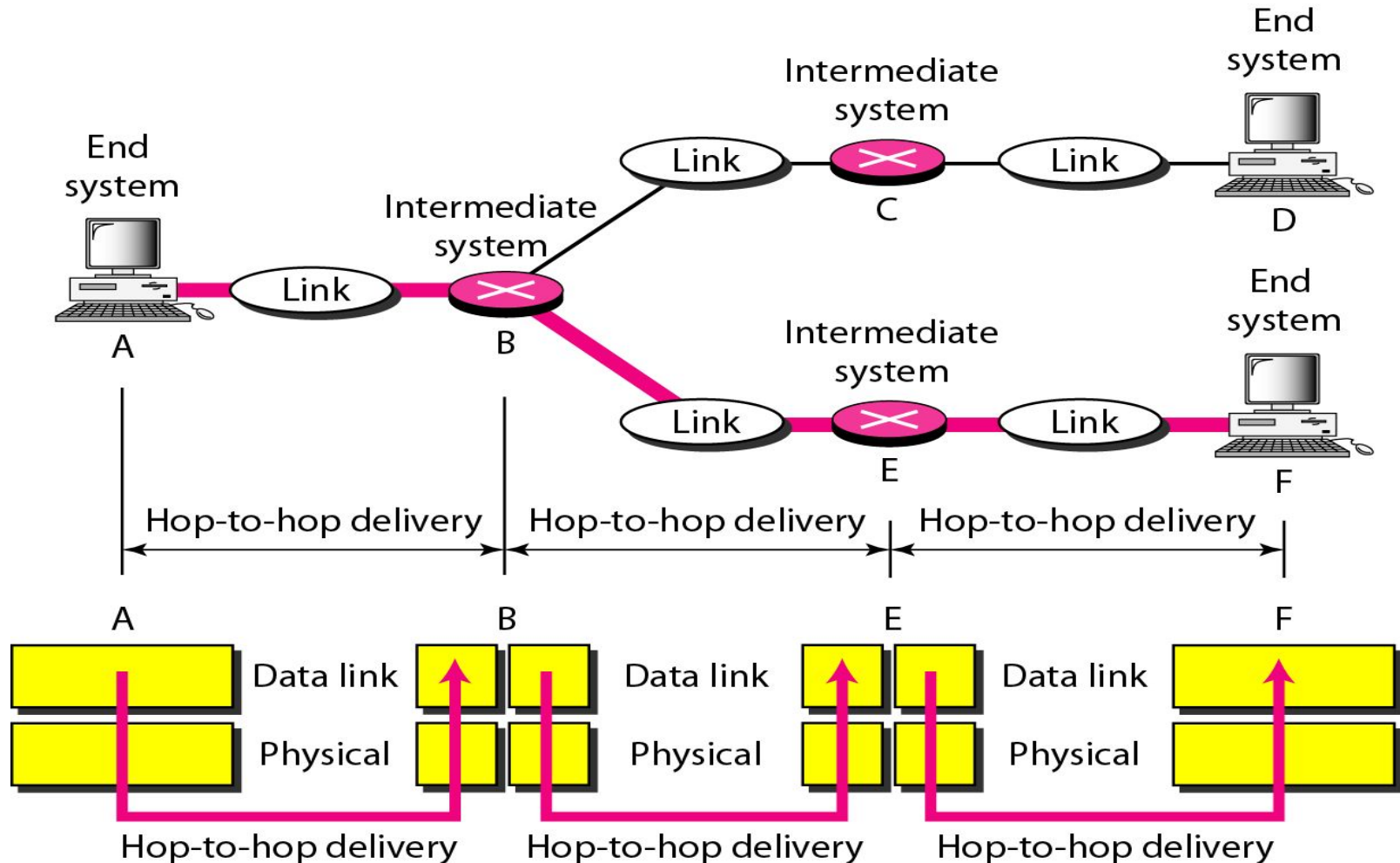


Figure 2.7 *Hop-to-hop delivery*

Physical Layer — Responsible for Bit Transmission

Function:

- The Physical Layer handles the actual physical connection between two devices (for example, cable, Wi-Fi signal, fiber).
- It transmits raw bits (0s and 1s) across this connection.

Role in Hop-to-Hop Delivery:

It provides the path that connects two adjacent nodes. It defines:

- The type of **transmission medium** (e.g., Ethernet cable, wireless signal).
- The **electrical/optical/radio signal** representation of bits.
- **Synchronization** and line **encoding**.

Without the physical layer, two devices cannot physically communicate.

Data Link Layer — Responsible for Reliable Frame Delivery

Function: It ensures that the bits sent over the physical connection are organized and delivered without errors.

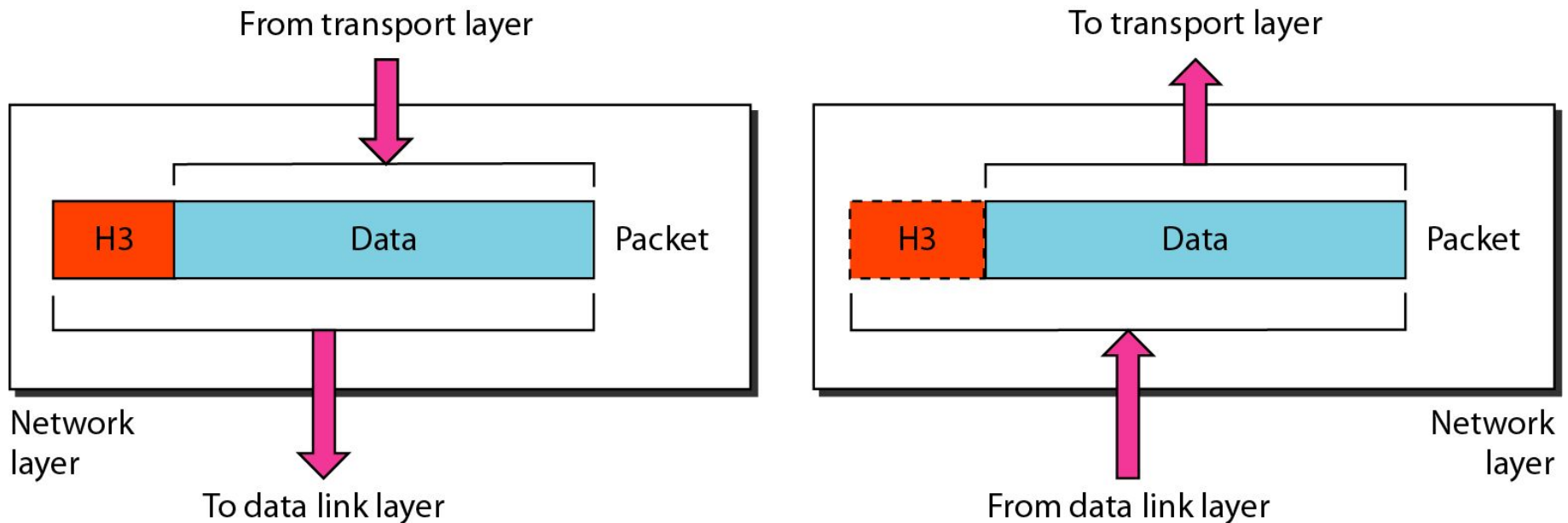
Role in Hop-to-Hop Delivery: It creates frames from network layer packets and adds Header (H2) and Trailer (T2). It ensures:

- Frames are sent to the correct next device.
 - Errors in transmission are detected and possibly corrected.
 - Retransmission happens if necessary (in reliable links).

The data link layer guarantees that the frame reaches the next node correctly over the physical link.

Figure 2.8 *Network layer*

- The Network Layer is responsible for end-to-end delivery of packets that means it makes sure data goes from the source device to the final destination device, possibly across many intermediate routers.
- So while the Data Link Layer handles **hop-to-hop** delivery, the Network Layer handles **source-to-destination** (end-to-end) delivery.



Working at the Sender Side (Encapsulation)

1. Receives Data from Transport Layer:

The Transport Layer gives a segment (data + header) to the Network Layer.

2. Adds a Network Header (H3): The Network Layer adds its own header (H3) that includes:

- Source IP address
- Destination IP address
- Routing information
- Protocol info (TCP/UDP, etc.)

After adding this header, the data unit becomes a **packet** (H3 + Data).

3. Determines Best Path: The Network Layer uses routing algorithms to decide which path the packet should take through the network to reach the destination.

4. Passes Packet to the Data Link Layer: The packet is then passed down to the Data Link Layer, which will handle hop-to-hop delivery along that chosen path.

Working at the Receiver Side (De-encapsulation)

1.Receives Packet from Data Link Layer

- The Data Link Layer gives the packet to the Network Layer after verifying it was received correctly.

2.Checks Destination IP

- The Network Layer checks if the destination IP address in the header matches the device's own IP.
- If it's the final destination → it keeps the packet.
- If it's not the final destination → it forwards the packet to the next hop based on its routing table.

3.Removes Header (H3)

- Once the packet reaches its destination, the Network Layer removes its header and passes the data up to the Transport Layer.

Note

The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Figure 2.9 *Source-to-destination delivery*

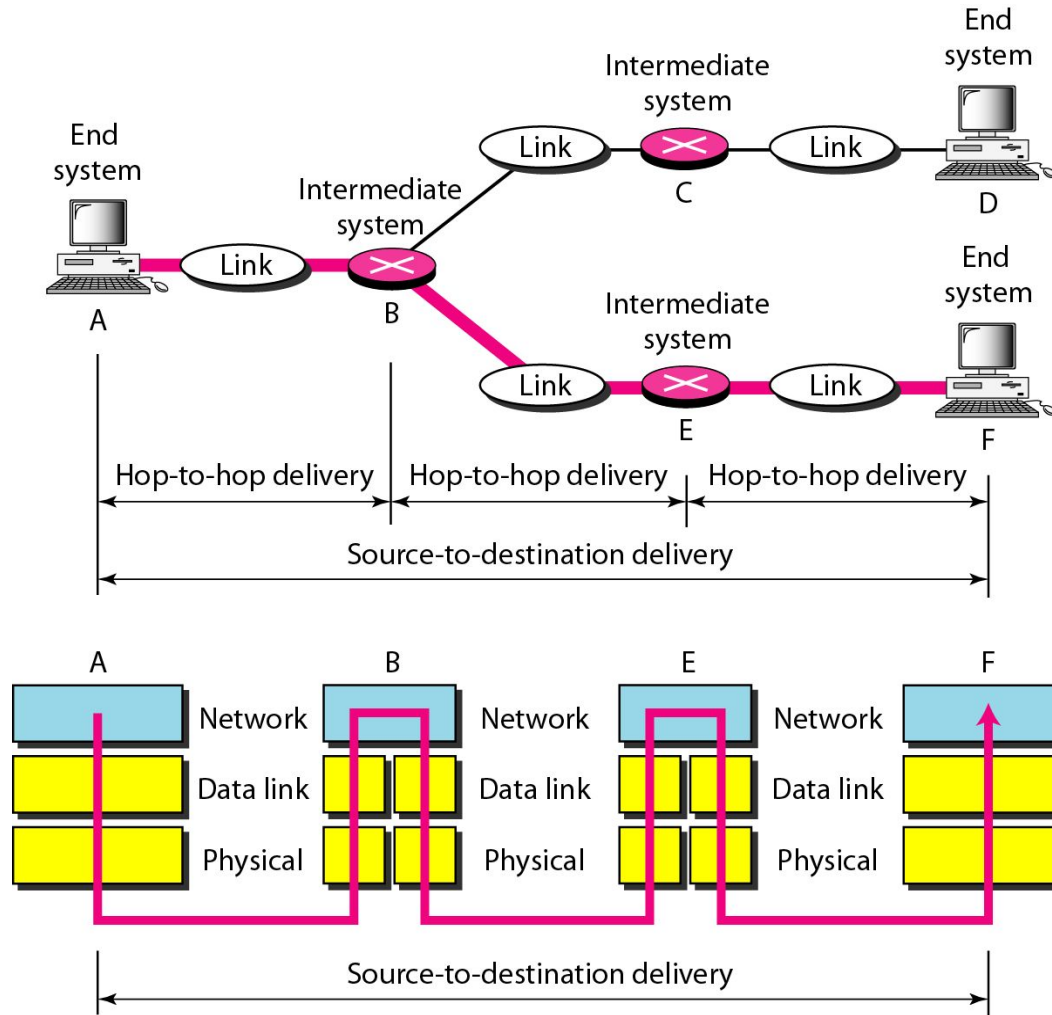
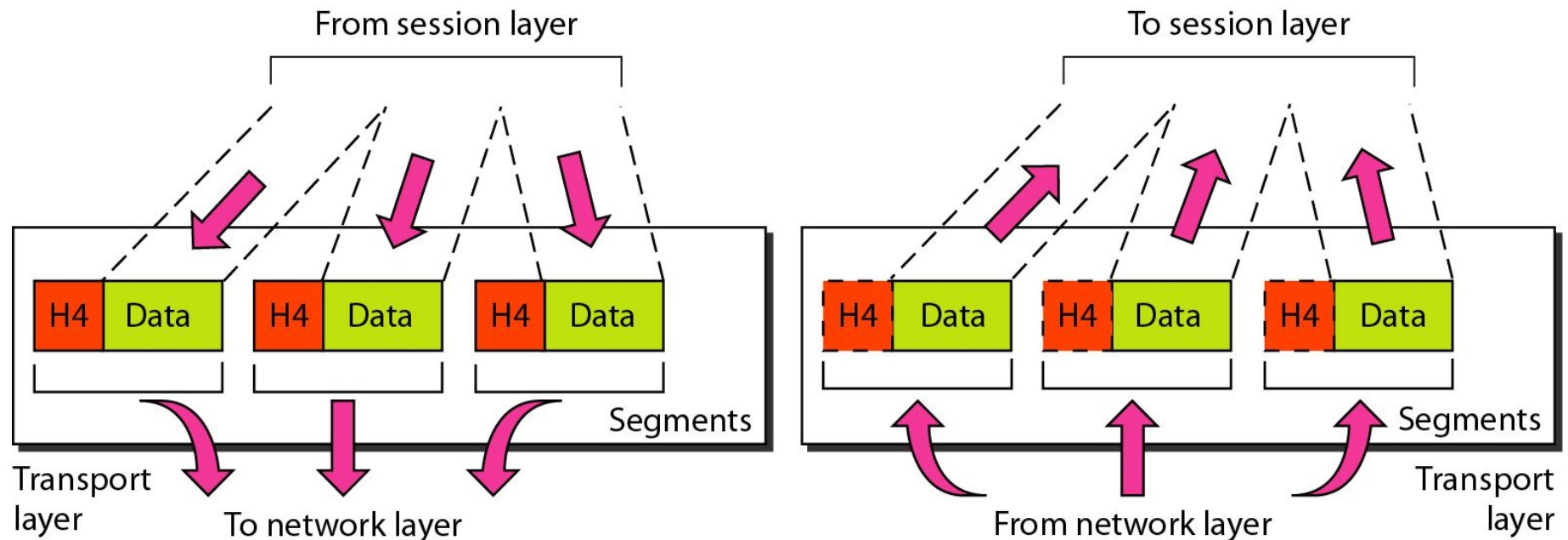


Figure 2.10 *Transport layer*

- The Transport Layer provides process-to-process delivery — meaning it delivers data from one specific application on the sender's computer to the correct application on the receiver's computer.
- **“Transport Layer makes sure your message reaches the correct app not just the correct computer.”**



Transport layer

Working at the Sender Side (Encapsulation)

1.Receives Data from the Session Layer

- The Session Layer sends data (message or stream) to the Transport Layer.

2.Segmentation

1. The data is broken into smaller segments (if large) to fit network limits.

4.Adds Transport Header (H4)

Each segment gets a Transport Header, which includes:

- 1) Source Port Number** → identifies the sending application.
- 2) Destination Port Number** → identifies the receiving application.
- 3) Sequence Number** → helps reassemble data in correct order.
- 4) Acknowledgment Number** → confirms successful receipt.
- 5) Flags & Control Bits** → manage connection setup, termination, and flow.

Data unit at this layer is called a **Segment**. A segment in networking is the data unit at the transport layer, especially in TCP, containing part of the message plus control information (like port numbers, sequence numbers, etc.).

4. Passes Segments to the Network Layer

1. These segments are then sent down to the Network Layer, which will handle end-to-end routing between devices

Figure 2.10 *Transport layer*

Working at the Receiver Side (De-encapsulation)

1. Receives Segments from Network Layer

- The Transport Layer collects incoming packets, extracts the segment, and reads the header.

2. Error Checking & Acknowledgment

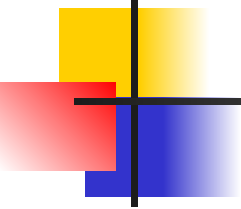
- 1) It checks for errors or missing segments.
- 2) If using TCP, it requests retransmission of lost segments.
- 3) Sends acknowledgments (ACKs) for successfully received segments.

4. Reassembly

- Using sequence numbers, it reassembles all segments in the correct order.

5. Delivers Data to the Correct Application

- 1) Using port numbers, it identifies which application (like Chrome or Gmail) should receive the data.
- 2) Then it passes the reassembled data up to the Session Layer (or directly to the application).



Note

The transport layer is responsible for the delivery of a message from one process to another.

Figure 2.11 *Reliable process-to-process delivery of a message*

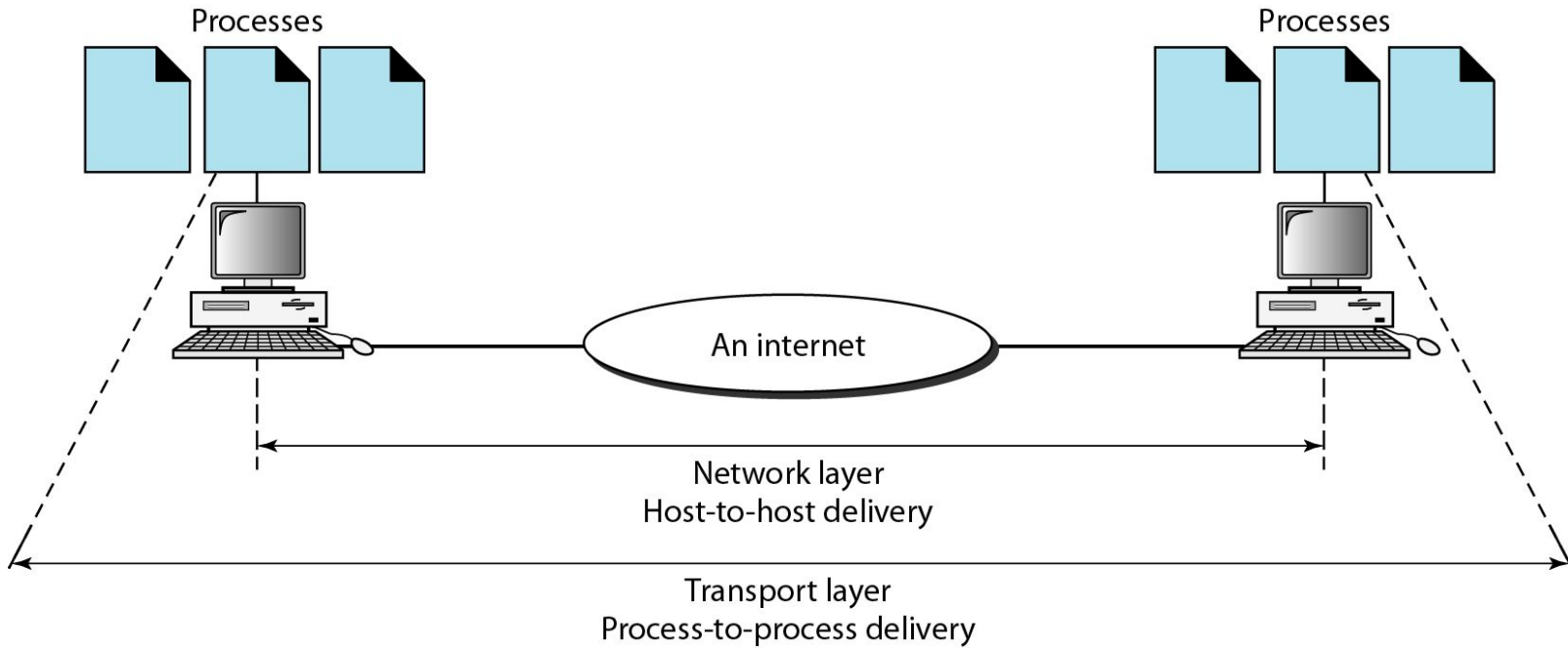


Figure 2.12 *Session layer*

- The Session Layer is responsible for establishing, managing, and ending sessions (connections) between two communicating devices or applications.

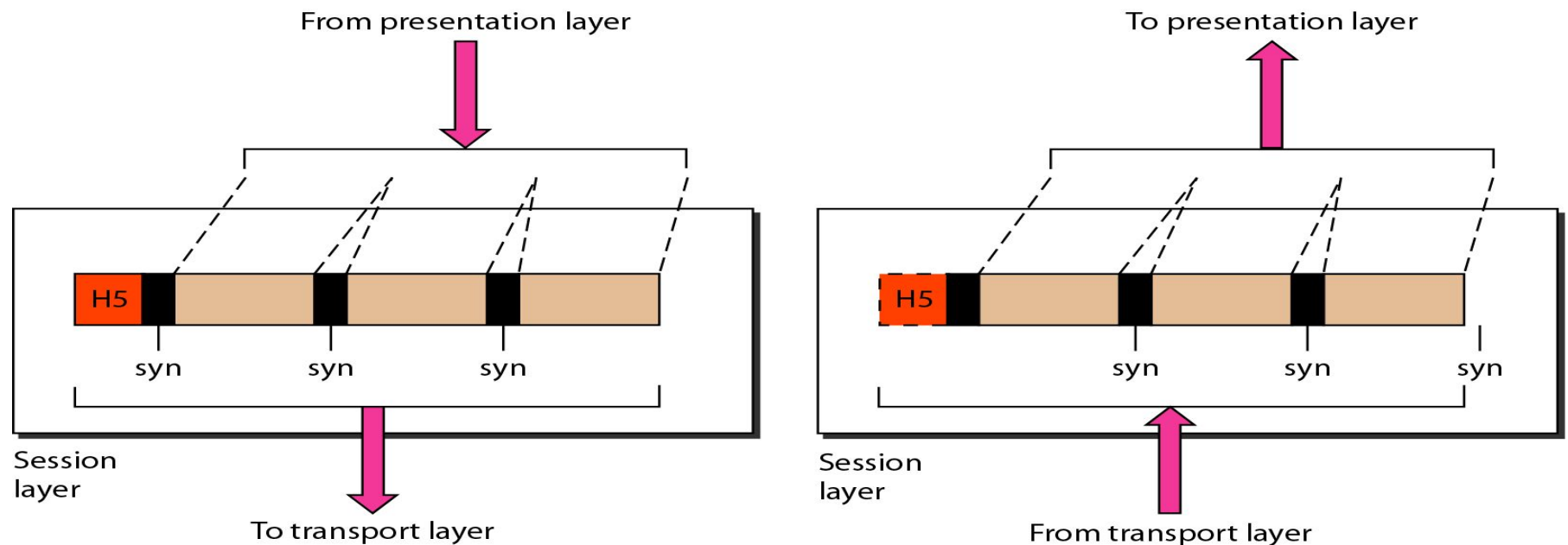


Figure 2.12 *Session layer*

Sender Side:

1. Session Establishment

- Before two applications start exchanging data, the session layer creates a session (a communication link).
- It decides who will speak first, and how long each side can talk (dialog control).

2. Session Maintenance

- Keeps the session active while data is being exchanged.
- Ensures that if something goes wrong (like a temporary disconnection), the session can resume instead of starting over.

3. Synchronization

- Adds checkpoints (synchronization points) during long transmissions.
- So, if a failure happens, transmission can restart from the last checkpoint, not from the beginning.

4. Passing Data to Transport Layer (Sender Side)

- After adding the H5 header and synchronization points, the Session Layer passes the data down to the Transport Layer.
- The Transport Layer will then segment and transmit it over the network.

Figure 2.12 *Session layer*

Receiver Side:

- The Transport Layer at the receiver end delivers data up to the Session Layer.
- The Session Layer checks the H5 header and identifies synchronization points.
- Using these points, it ensures the session is correctly reassembled and in order.
- Finally, the Session Layer sends the cleaned and reassembled data up to the Presentation Layer.

2.Session Management

- Accepts and maintains the session from the sender.
- Keeps track of data exchange who is sending and who is receiving.

3.Session Termination

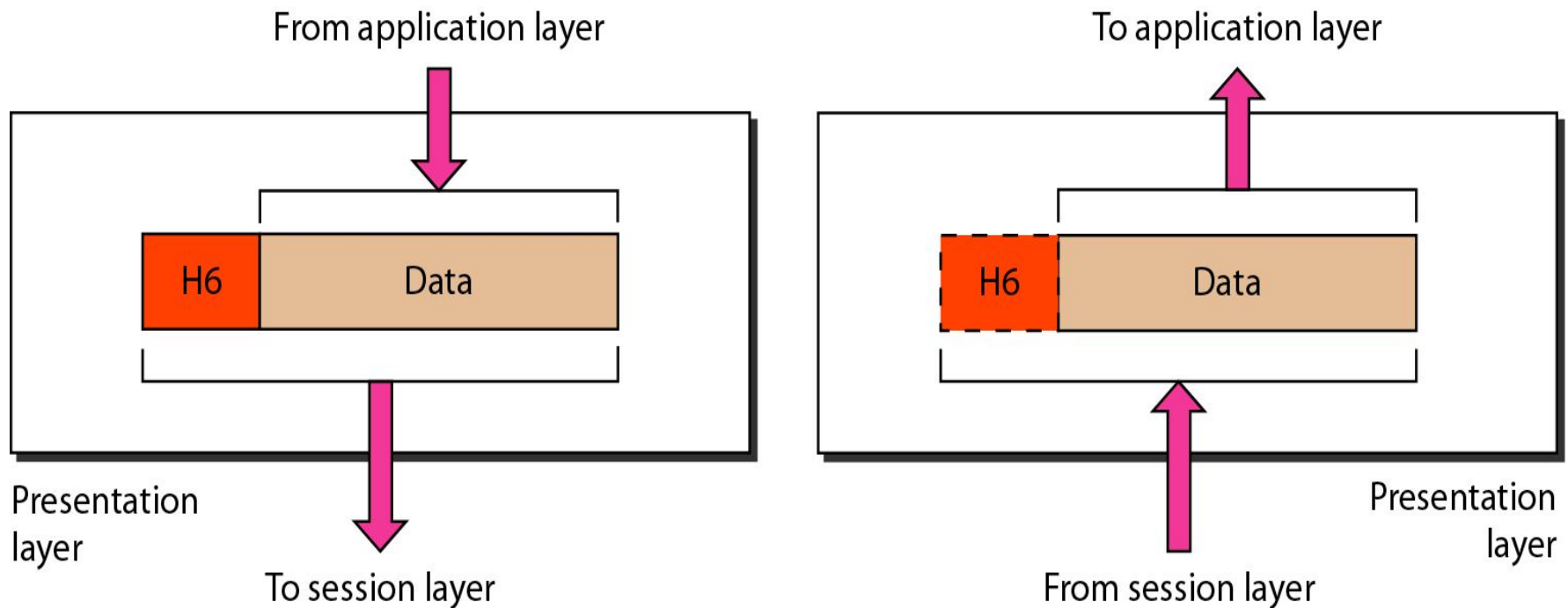
- After communication is complete, the session layer gracefully closes the session to free resources.

Note

The session layer is responsible for dialog control and synchronization.

Figure 2.13 *Presentation layer*

- The Presentation Layer is responsible for the translation, encryption/decryption, and compression/decompression of data between the application layer and the lower layers.
- It makes sure the data from one system can be understood by another system no matter their format or architecture.



Presentation layer

Sender Side:

- The Application Layer sends raw data down to the Presentation Layer.
- The Presentation Layer adds its own header (H6), which contains information about data formatting, encryption, and compression.
- After processing, it sends the formatted data down to the Session Layer.

1. Translation

- Converts the data from the application format into a standard network format that can be transmitted.
- For example: text, images, or videos are converted into a format that all systems can understand.

2. Encryption

- Data is encrypted (secured) before being sent, so unauthorized users cannot read it.
- Example: converting text “HELLO” into unreadable code like “X5#7!A”.

3. Compression

- Reduces the size of the data for faster transmission.
- Example: compressing images or videos before sending

Presentation layer

Receiver Side:

- The Presentation Layer receives the data from the Session Layer. It removes the H6 header and performs reverse operations — decryption, decompression, and translation — before sending it up to the Application Layer.

1.Decompression

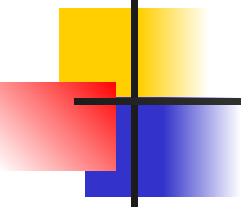
- Expands the compressed data back to its original size.

2. Decryption

- Converts the encrypted data back into readable form using the correct key.

3. Translation

- Converts the standardized network data back into a format that the application can understand (like turning binary data into readable text or images).



Note

The presentation layer is responsible for translation, compression, and encryption.

Figure 2.14 *Application layer*

- The Application Layer provides the interface between the user (or software applications) and the network.
- It's the layer where you actually interact with the network — through apps like web browsers, email clients, or chat software.

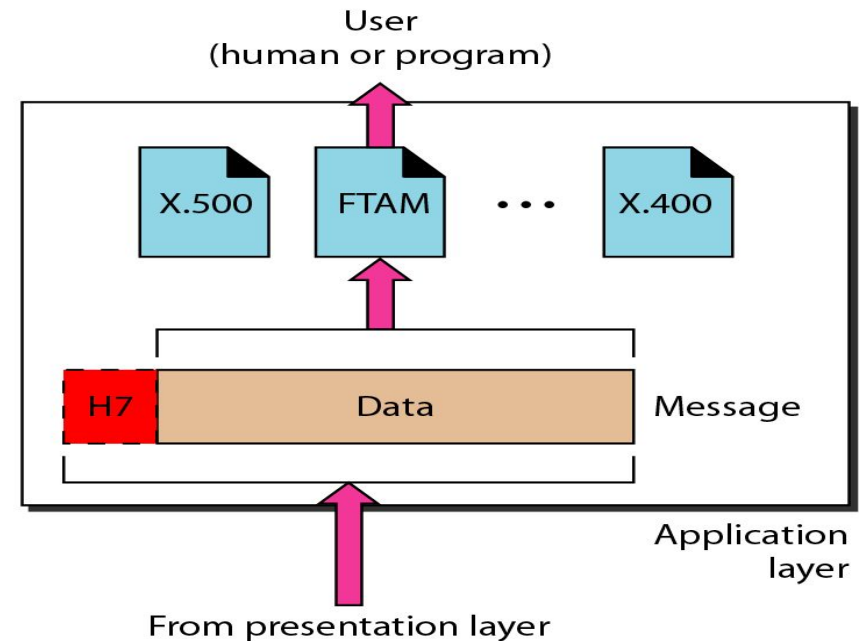
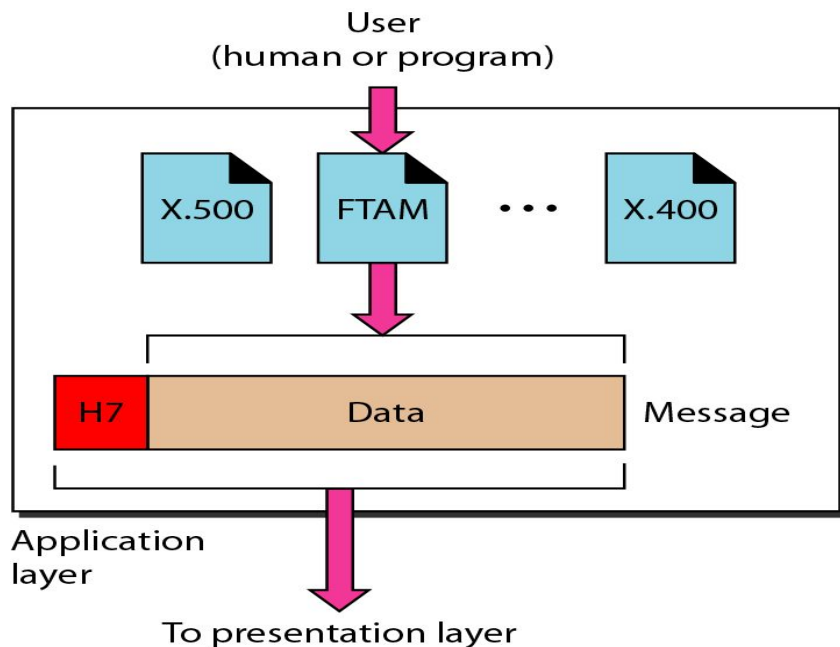


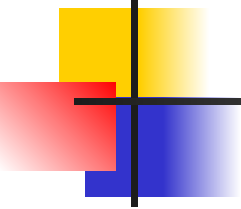
Figure 2.14 *Application layer*

At Sender Side

- A user or software wants to send data (e.g., you send an email using Gmail). The Application Layer prepares the data for transmission by:
 - Choosing the right protocol (like HTTP for web, SMTP for email). Passing the data to the Presentation Layer for formatting and encryption.
 - **The Application Layer:**
 - Adds H7 header (defines service and message type).
 - Checks that the receiver and resources are ready.
 - Sends the complete message to the Presentation Layer.
- Think of it as: "The layer where actual user tasks like sending email, browsing, or transferring files begin."

At Receiver Side

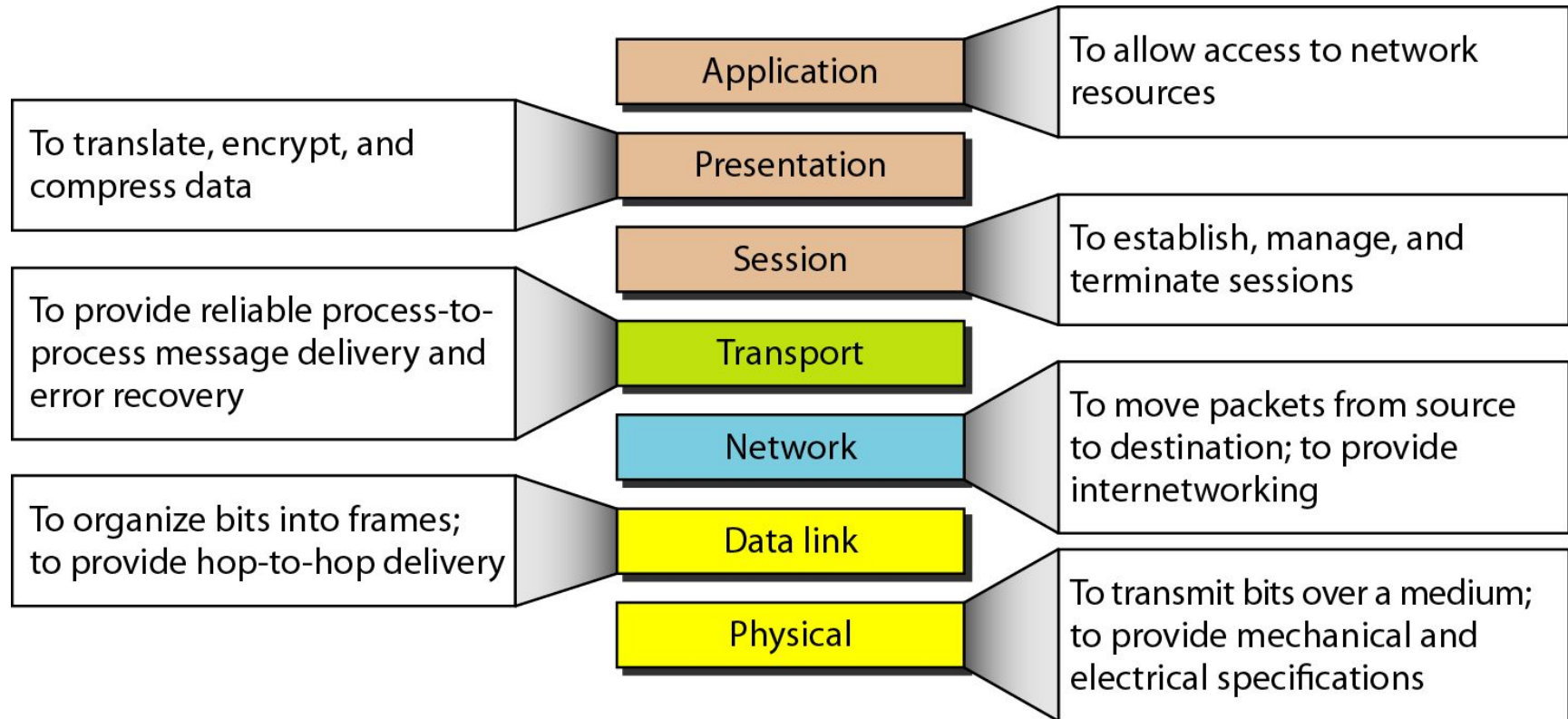
- Receives message from Presentation Layer.
- Removes H7 header.
- Passes the message to the correct application or user interface (like email inbox, file manager, etc.).



Note

The application layer is responsible for providing services to the user.

Figure 2.15 *Summary of layers*



2-4 TCP/IP PROTOCOL SUITE

*The layers in the **TCP/IP protocol suite** do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers: **host-to-network**, **internet**, **transport**, and **application**. However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: **physical**, **data link**, **network**, **transport**, and **application**.*

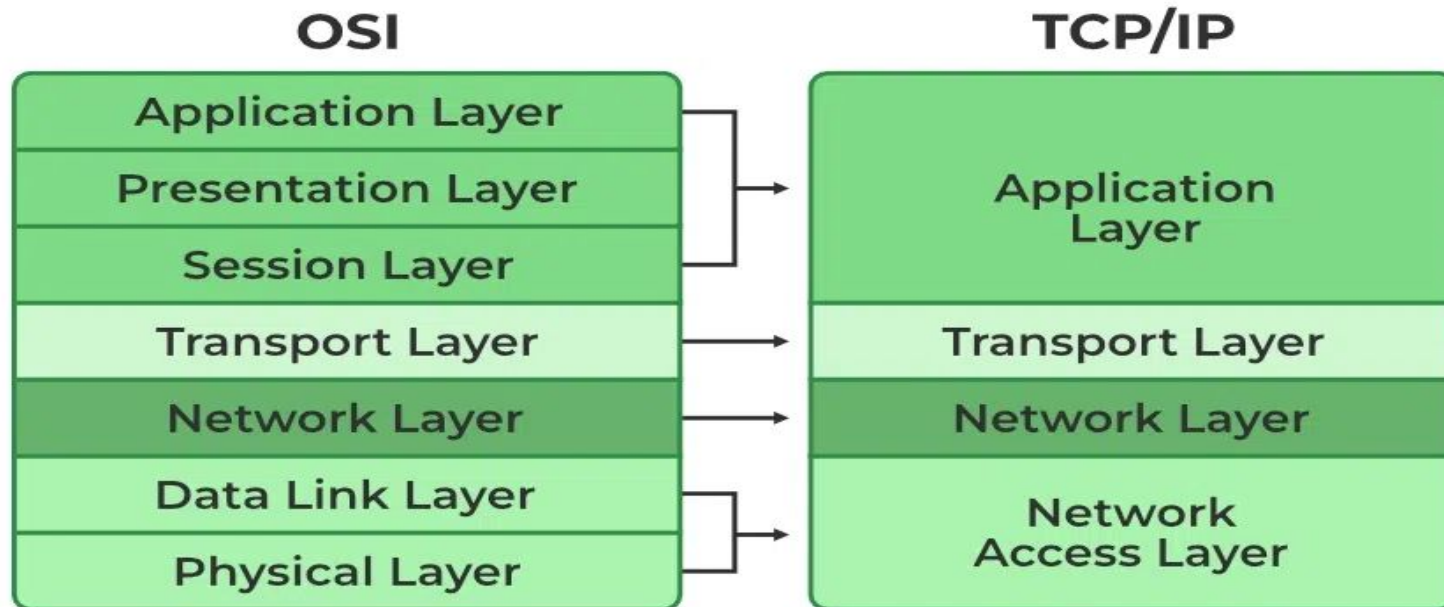
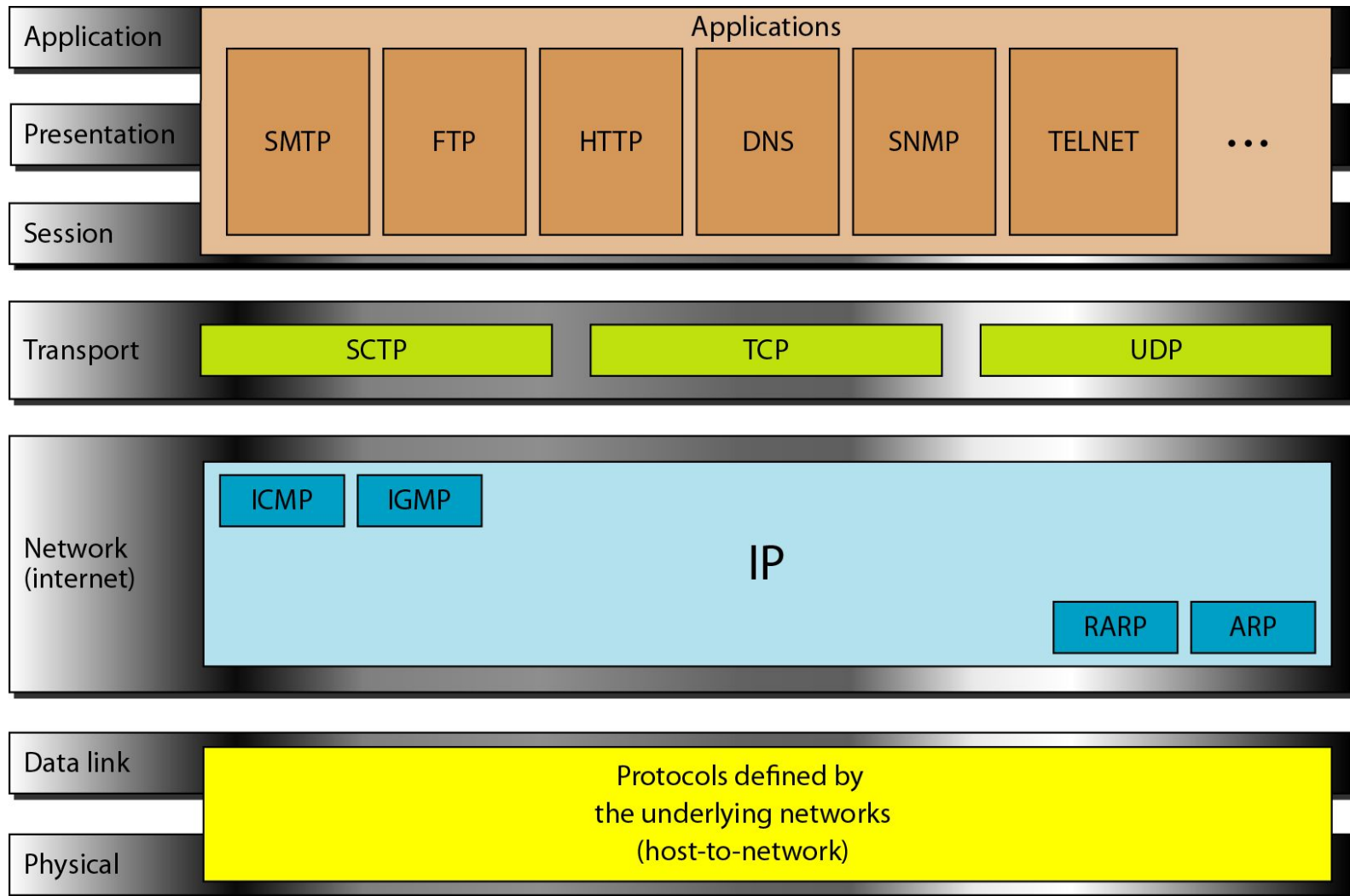


Figure 2.16 *TCP/IP and OSI model*



1. Application Layer

•Includes:

- **SMTP** (Simple Mail Transfer Protocol) – for email
- **FTP** (File Transfer Protocol) – for file transfer
- **HTTP** (Hypertext Transfer Protocol) – for web browsing
- **DNS** (Domain Name System) – for converting domain names to IP addresses
- **SNMP** (Simple Network Management Protocol) – for managing network devices
- **TELNET** – for remote login

In the TCP/IP model, the Application Layer combines the OSI model's **Application, **Presentation**, and **Session** layers.**

Transport Layer

Protocols:

- **TCP** (Transmission Control Protocol) – reliable, connection-oriented communication (used by HTTP, FTP, etc.)
- **UDP** (User Datagram Protocol) – fast, connectionless communication (used by DNS, streaming, etc.)
- **SCTP** (Stream Control Transmission Protocol) – used for applications needing multiple streams of data.

Network Layer ↔ (Internet Layer):

Supporting Protocols:

- **ICMP** (Internet Control Message Protocol) – for error reporting and diagnostics.
 - **IGMP** (Internet Group Management Protocol) – for managing multicast group memberships.
 - **ARP** (Address Resolution Protocol) – maps IP addresses to MAC (hardware) addresses.
 - **RARP** (Reverse ARP) – maps MAC addresses to IP addresses.
-

Data Link & Physical Layers (Host-to-Network Layer)

Purpose:

- Deals with the hardware-level transmission of data over physical media (like cables or wireless).

Includes:

- Protocols defined by the underlying network technology (e.g., Ethernet, Wi-Fi).
- This layer ensures hop-to-hop delivery between directly connected nodes.

2-5 ADDRESSING

*Four levels of addresses are used in an internet employing the TCP/IP protocols: **physical**, **logical**, **port**, and **specific**.*

Topics discussed in this section:

Physical Addresses

Logical Addresses

Port Addresses

Specific Addresses

Figure 2.17 *Addresses in TCP/IP*

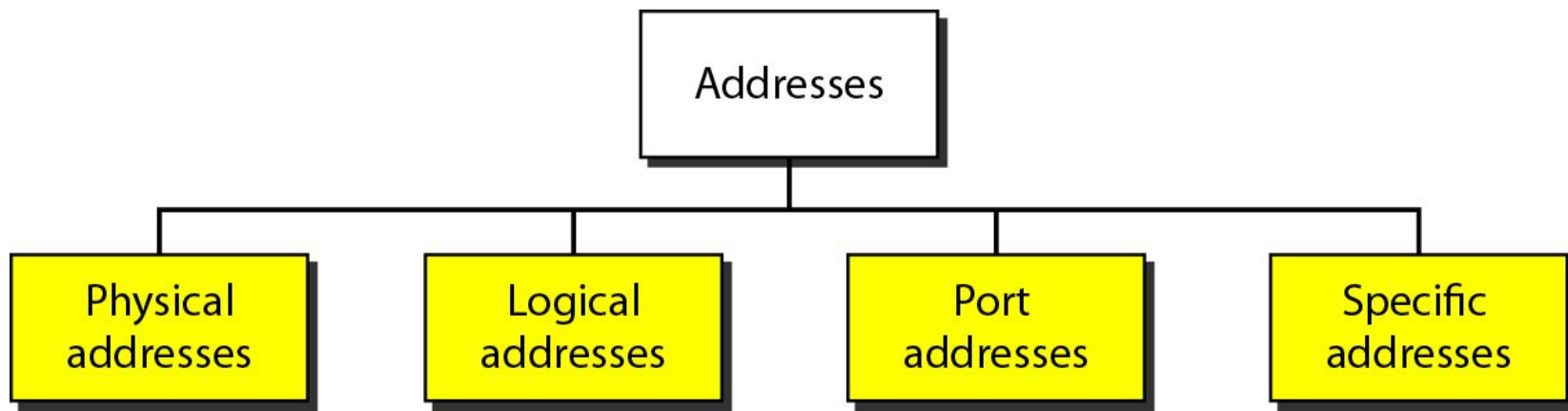


Figure 2.18 *Relationship of layers and addresses in TCP/IP*

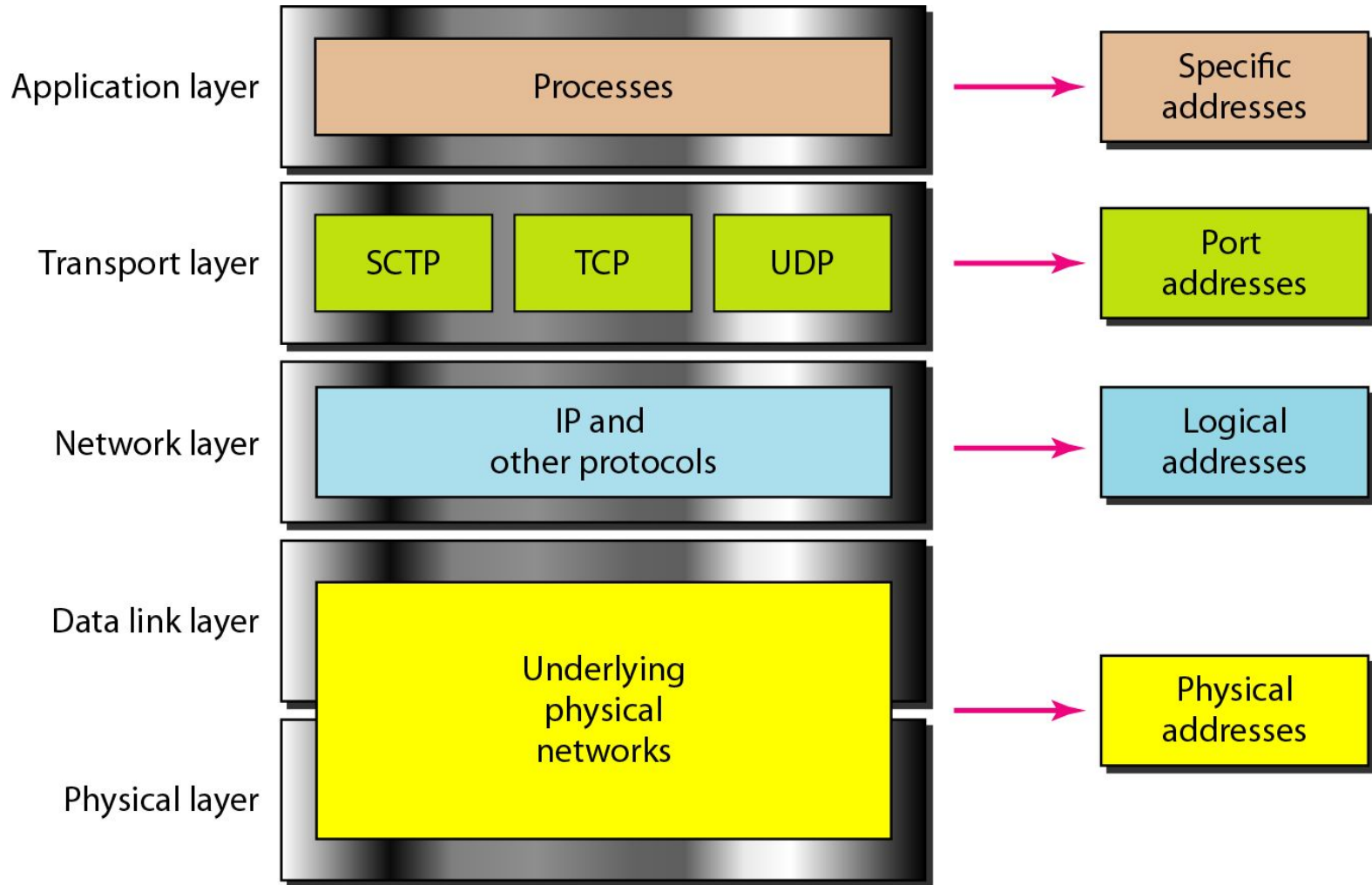
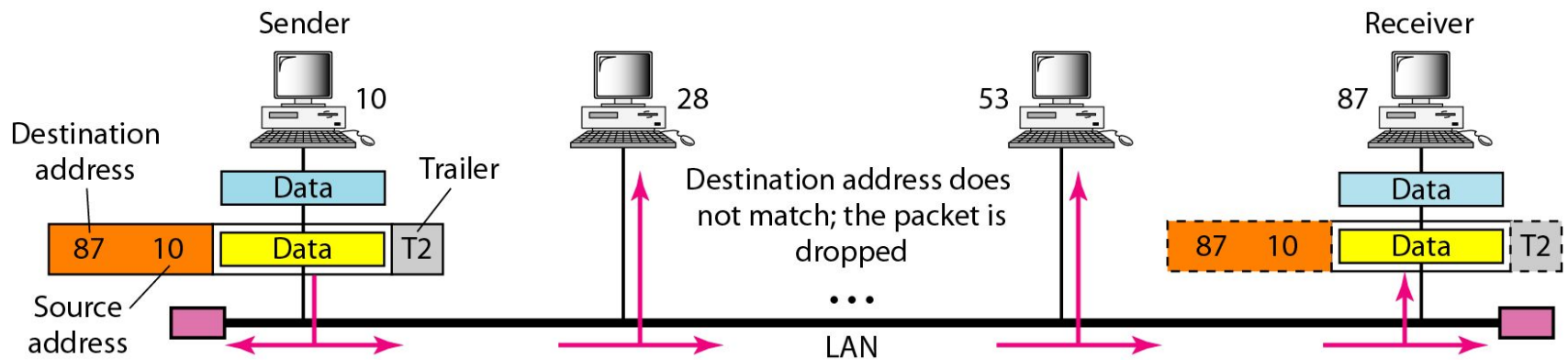


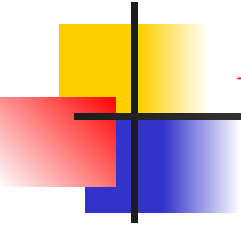
Figure 2.18 *Relationship of layers and addresses in TCP/IP*

Layer	Protocol Examples	Type of Address	Example	Purpose
Application	HTTP, SMTP, DNS	Specific Address	www.google.com	Identify the application/service
Transport	TCP, UDP, SCTP	Port Address	Port 80, Port 53	Identify process on a host
Network (Internet)	IP, ICMP, IGMP	Logical Address	192.168.1.10	Identify host location on the network
Data Link / Physical	Ethernet, Wi-Fi	Physical Address (MAC)	00:1A:2B:3C:4D:5E	Identify device's hardware interface

Figure 2.19 Physical addresses

*In Figure 2.19 a node with physical address 10 sends a frame to a node with physical address 87. The two nodes are connected by a link (bus topology LAN). As the figure shows, the computer with physical address **10** is the sender, and the computer with physical address **87** is the receiver.*



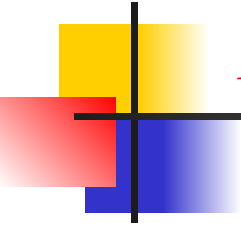


Example 2.2

*As we will see in Chapter 13, most local-area networks use a **48-bit** (6-byte) physical address written as 12 hexadecimal digits; every byte (2 hexadecimal digits) is separated by a colon, as shown below:*

07:01:02:01:2C:4B

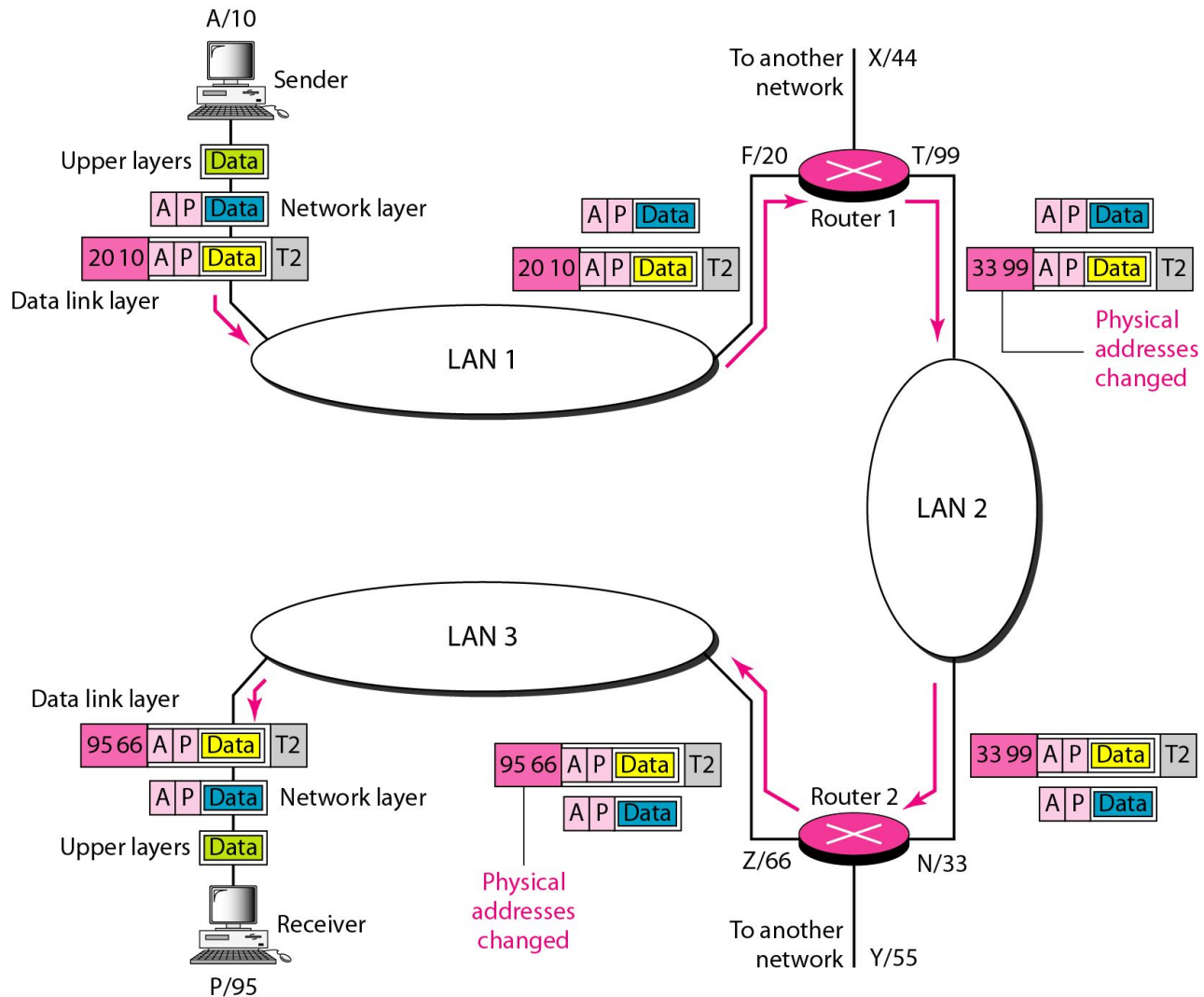
A 6-byte (12 hexadecimal digits) physical address.

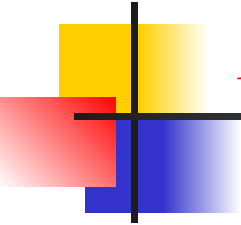


Example 2.3

Figure 2.20 shows a part of an internet with two routers connecting three LANs. Each device (computer or router) has a pair of addresses (logical and physical) for each connection. In this case, each computer is connected to only one link and therefore has only one pair of addresses. Each router, however, is connected to three networks (only two are shown in the figure). So each router has three pairs of addresses, one for each connection.

Figure 2.20 *IP addresses*

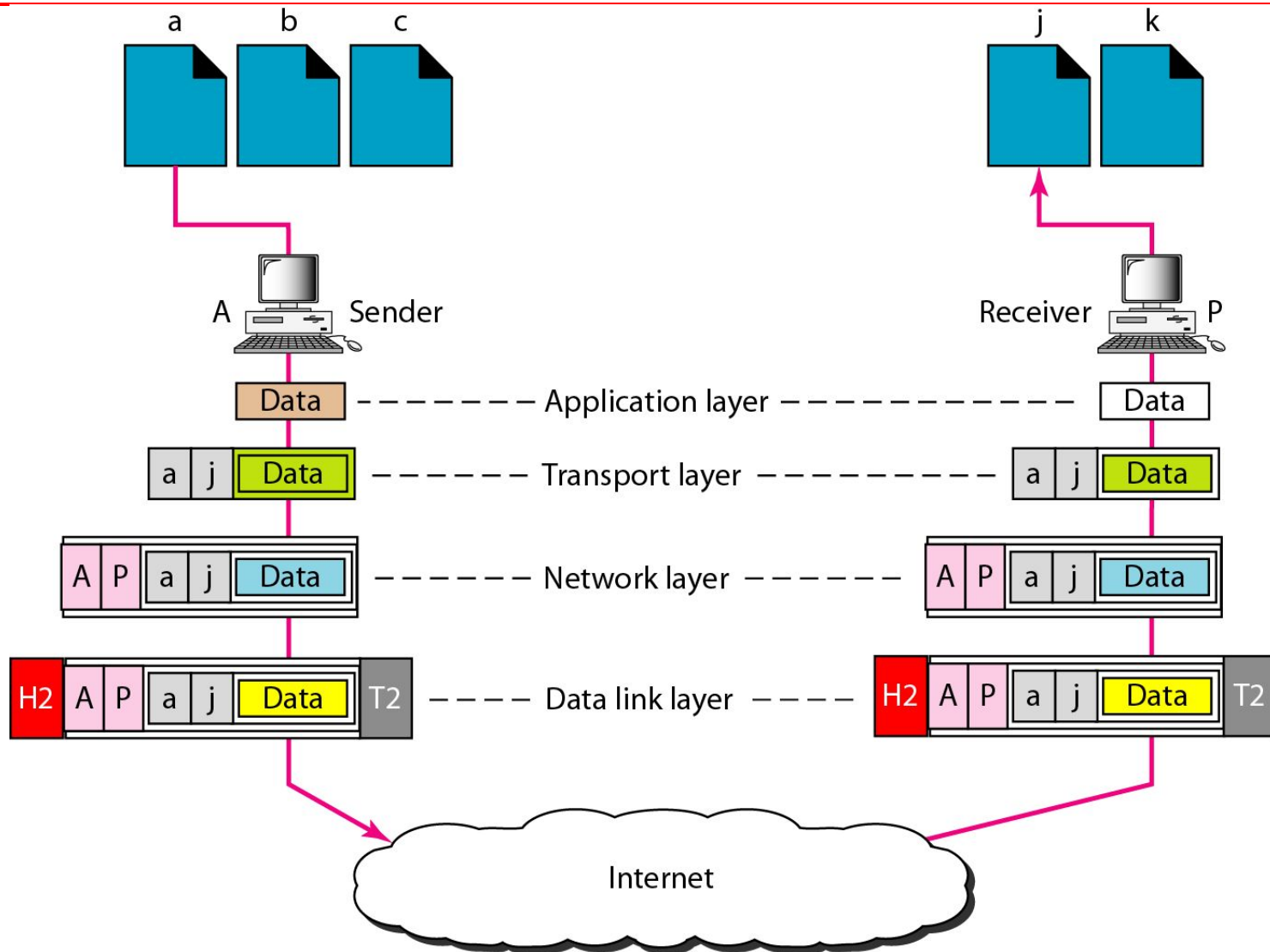




Example 2.4

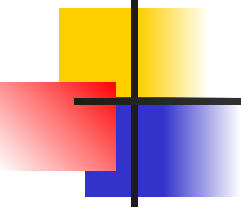
Figure 2.21 shows two computers communicating via the Internet. The sending computer is running three processes at this time with port addresses a , b , and c . The receiving computer is running two processes at this time with port addresses j and k . Process a in the sending computer needs to communicate with process j in the receiving computer. Note that although Physical addresses change from hop to hop, Logical and port addresses remain the same from the source to destination.

Figure 2.21 *Port addresses*



Note

**The physical addresses change from hop to hop,
but the logical and port addresses usually remain the same.**



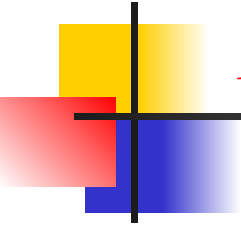
Physical Address (MAC Address) → Changes Hop to Hop

- The physical address (MAC address) belongs to the Network Interface Card (NIC).When data travels through the network, it passes through different devices (routers, switches, etc.).At each hop, the sender and receiver's MAC addresses change, because every link has its own physical network.

Example: Imagine a message going from Computer A → Router → Router → Computer B.

Link	Sender MAC	Receiver MAC
A → Router1	A's MAC	Router1's MAC
Router1 → Router2	Router1's MAC	Router2's MAC
Router2 → B	Router2's MAC	B's MAC

So, MAC address changes at every step (hop).It only helps deliver data from one device to the next directly connected device.



Example 2.5

As we will see in Chapter 23, a port address is a 16-bit address represented by one decimal number as shown.

753

**A 16-bit port address represented
as one single number.**